

Second-Order Linear ODE's

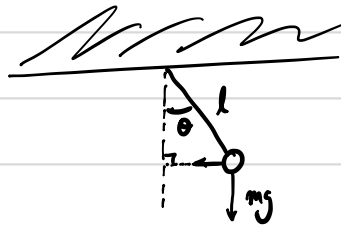
Newton's Second Law states: $m \frac{dv}{dt} = \sum_i F_i$, where F_i is some external force.

or more commonly: $m \ddot{x} = \sum_i F_i$, in some inertial frame.

Below are examples of some external forces we sometimes encounter in practice:

① Restoring Forces

Pendulum



Restoring force is given by $-mg \sin \theta$.

The ODE $\ddot{\theta} = -\frac{g}{l} \sin \theta$ is too difficult to solve, so we make a small-angle approximation $\sin \theta \approx \theta$ for small θ .

$$\text{Then } \ddot{\theta} = -\frac{g}{l} \theta$$

Mass-Spring Systems

Mass-Spring Systems follow a simplified linear restoring force as well, for small displacements:

$$m \ddot{x} = -kx$$

LC-Circuits

$$L \ddot{I} = -\frac{1}{C} I$$

L : tries to resist change

C : tries to expel stored energy

Current bounces back and forth w/in the circuit.



② Damping Forces

Mechanical

$$F_{\text{damp}} = -bv$$

or

$$m\ddot{x} = -b\dot{x}$$

Consider WS-02 examples involving linear wind-resistance examples. Wind-resistance acts like a friction force proportional to speed.

LR-Circuits

$$L\ddot{I} = -RI$$

An LR-Circuit is similar. Inductors resist change whereas resistors create voltage drops. Resistors, here, act like frictional forces proportional to current rate of change.

③ "other" forces: Both mechanical & electrical

$F = F_{\text{ext}} = f(t)$ some forces dependent on time. Related to amplifiers, or other (possibly) human factors.

This acts like a "catch-all" term.

Applying Newton's Second Law to all of these cases and we get:

$$m\ddot{x} = -kx - b\dot{x} + f(t) \quad \text{or} \quad m\ddot{x} + b\dot{x} + kx = f(t)$$

This is a linear Second-order ODE!

The linear operator is given by $L(x) = f(t)$

$$\text{where } L(\cdot) = \left(m \frac{d^2}{dt^2} + b \frac{d}{dt} + k \right) (\cdot) = f(t)$$

$$\text{"A"} \quad \text{"x"} = \text{"b"}$$

Any solution that satisfies the above equation is an example of Simple Harmonic Motion.

Defn: (S.H.O.) Simple Harmonic Oscillator

The simple harmonic oscillator equation is

$$m\ddot{x} + b\dot{x} + kx = f(t),$$

a second-order nonhomogeneous linear ODE w/ constant coefficients.

$$m > 0, \quad b \geq 0, \quad k > 0.$$

If $b=0$ we say the [physical] system is undamped. Otherwise it is damped.

If $f(t)=0$ the equation is homogeneous, or unforced, undriven, free.
Otherwise it is forced, driven.

Important Side Note: Generalization of Picard's Theorem for Existence & Uniqueness

Let $p(t)$ & $q(t)$ be continuous on the open interval (a,b) containing t_0 .
For ANY A & B in $\mathbb{R}$, there exists a unique solution $y(t)$ defined on (a,b) to the IVP

$$y'' + p(t)y' + q(t)y = 0$$

$$y(t_0) = A$$

$$y'(t_0) = B$$

(No proof, slightly beyond us)

Note: Since the system is linear, superposition applies. If y_1 & y_2 are solutions, $\alpha y_1 + \beta y_2$ is also a solution.

Theorem: The same applies to the N^{th} -Order Linear Homogeneous IVP.

$$y^{(n)}(t) + p_1(t)y^{(n-1)}(t) + \dots + p_n(t)y = 0$$

$$y(t_0) = A_0$$

$$y'(t_0) = A_1$$

$\vdots$

$$y^{(n-1)}(t_0) = A_{n-1}$$

The solution exists and is unique.

For more details, see p. 217-219 in book.

As a consequence: Corollary: The solution space of an n^{th} order ODE is n -dimensional.

Back to ODE's as descriptors to physical systems!

Note from Calculus, we already know that the eqn

$$\ddot{x} + x = 0$$

is solved by

$$x(t) = C_1 \cos(t) + C_2 \sin(t).$$

What if we said $x(t) = e^{\lambda t}$ for some λ ? Then $\dot{x} = \lambda e^{\lambda t}$, $\ddot{x} = \lambda^2 e^{\lambda t}$, so that

$$\ddot{x} + x = \lambda^2 e^{\lambda t} + e^{\lambda t} = (\lambda^2 + 1)e^{\lambda t} = 0$$

Since the exponential is never zero, $\lambda^2 + 1 = 0 \Rightarrow \underline{\lambda = \pm i}$

If e^{it} & e^{-it} solve the system, so does $x(t) = C_1 e^{it} + C_2 e^{-it}$!

But, from the existence & uniqueness theorem they must be the same solution!

$$\therefore A e^{it} + B e^{-it} = C \cos(t) + D \sin(t)$$

$$\begin{array}{l} t=0 \\ A+B=C \\ t=\pi \\ A e^{i\pi} + B e^{-i\pi} = -C \end{array}$$

$$\begin{array}{l} t=\pi/2 \\ A e^{i\pi/2} + B e^{-i\pi/2} = D \\ t=3\pi/2 \\ A e^{3i\pi/2} + B e^{-3i\pi/2} = -D \end{array}$$

Suppose $B=0$, add left two:

$$A + A e^{i\pi} = 0$$

$$A(1 + e^{i\pi}) = 0$$

$$\Rightarrow \boxed{e^{i\pi} = -1} \quad \text{"Euler's Identity"}$$

$$\text{This implies } \underline{e^{i\pi/2} = i} \quad \text{and} \quad \underline{e^{3i\pi/2} = -i}$$

Use these to rewrite all conditions above

$$A+B=C \quad iA - iB = D$$

$$\text{Again, set } B=0, \text{ we get } A=C = \frac{D}{i}$$

So $A e^{it} = A \cos(t) + iA \sin(t)$, or $\underbrace{e^{it} = \cos(t) + i \sin(t)}_{\text{"Euler's Equation"}}$

By symmetry, then $e^{-it} = \cos(t) - i \sin(t)$, and we can add and subtract these equations to get:

$$\cos(t) = \frac{e^{it} + e^{-it}}{2} \quad \& \quad \sin(t) = \frac{e^{it} - e^{-it}}{2i}$$

What we learned is we can use real or complex exponentials exclusively to solve ODEs! They are the same thing! There is nothing imaginary about imaginary numbers!

We will solve ALL Harmonic Oscillator situations this way and study the solutions case-by-case.

First assume a solution of the form $x(t) = e^{\lambda t}$, for $\lambda \in \mathbb{C}$. Then this means

$$\begin{aligned} m \ddot{x} + b \dot{x} + kx &= 0 \\ (m \lambda^2 + b \lambda + k) e^{\lambda t} &= 0 \\ \text{or} \end{aligned}$$

$$m \lambda^2 + b \lambda + k = 0$$

This is called the Characteristic Equation of the linear ODE.

$$\lambda = \frac{-b \pm \sqrt{b^2 - 4mk}}{2m}$$

← "roots of the characteristic equation"
or "eigenvalues"

This algebraic equation involving lambda will tell us much about physical systems.

Label $\Delta = b^2 - 4mk$ (the discriminant)

Then all solutions to the linear Second-Order ODE are solved by

$$x(t) = c_1 e^{\lambda_1 t} + c_2 e^{\lambda_2 t} = e^{\frac{-b}{2m} t} \left(c_1 e^{\frac{\sqrt{\Delta}}{2m} t} + c_2 e^{-\frac{\sqrt{\Delta}}{2m} t} \right)$$

Recall from elementary Algebra, we get three distinct solutions to the characteristic polynomial depending on the behavior of Δ .

If $\Delta > 0$: Two real distinct roots.
 $\Delta = 0$: One real repeated root.
 $\Delta < 0$: Two complex roots.

We will study these one at a time, last one first. Before we do that, consider the linear system we just studied in light of §3.6 material.

$x(t) = c_1 e^{\lambda_1 t} + c_2 e^{\lambda_2 t}$ is linear combination of vectors that span a 2D space. Since the solution space is 2D, and if $\lambda_1 \neq \lambda_2$, the exponentials $e^{\lambda_1 t}$ and $e^{\lambda_2 t}$ form a basis to the ODE!

Further note: Book defines $\alpha = \frac{-b}{2m}$ & $\beta = \frac{\sqrt{-\Delta}}{2m}$, so that

$$\alpha \pm i\beta = \lambda_{1,2}$$

Case 3, first: $\Delta < 0$.

Sub case(i): $b=0$ (Undamped Motion)

$$m\ddot{x} + kx = 0$$

The solutions take the form $x(t) = \tilde{C}_1 \cos(\omega_0 t) + \tilde{C}_2 \sin(\omega_0 t)$, where $\omega_0 = \sqrt{\frac{k}{m}}$

Recall that $\cos(A-B) = \sin(B) \cos(A) + \cos(B) \sin(A)$ (written that way on purpose)

We can write $x(t) = \tilde{C}_1 \cos(\omega_0 t) + \tilde{C}_2 \sin(\omega_0 t) = A \cos(\omega_0 t - \delta)$

A is the amplitude of the oscillation. $\left. \begin{array}{l} A \text{ is the amplitude of the oscillation.} \\ \delta \text{ is the phase angle (rad)} \end{array} \right\} \text{determined by I.C.s}$

ω_0 is the circular frequency or angular frequency. $\omega_0 = \sqrt{\frac{k}{m}}$ (rad/sec)

$f_0 = \omega_0 / 2\pi$ is the natural frequency or just frequency. ($\frac{1}{\text{sec}} = \text{Hz}$)

T is the period (sec).

$$T = \frac{1}{f_0} = \frac{2\pi}{\omega_0} = 2\pi \sqrt{\frac{m}{k}}$$

Thus, $x(t) = A \cos(\omega_0 t - \delta)$ is a horizontal shift of $A \cos(\omega_0 t)$ by $\frac{\delta}{\omega_0}$ (phase shift)

From the trigonometric identity, then,

$$\underline{\tilde{C}_1 = A \cos(\delta)} \quad \underline{\tilde{C}_2 = A \sin(\delta)}$$

$$\Rightarrow \underline{A^2 = \tilde{C}_1^2 + \tilde{C}_2^2} \quad \& \quad \underline{\tan(\delta) = \frac{\tilde{C}_2}{\tilde{C}_1}}$$

Ex/ A 5 kg mass hangs from the ceiling by a spring w/ constant $k = 20 \frac{N}{m}$. It is initially at rest in the equilibrium position. Suppose I kick it, giving it a vertical velocity of $1 \frac{m}{s}$. Find the position of the mass for all time.

$$5\ddot{x} + 20x = 0 \quad w/ \quad x(0) = 0, \quad \dot{x}(0) = 1$$

Solution is $x(t) = \tilde{C}_1 \cos(2t) + \tilde{C}_2 \sin(2t)$. Apply the I.C.s

$$x(0) = 0 = \tilde{C}_1 \cos(2(0)) + \tilde{C}_2 \sin(2(0)) = \tilde{C}_1$$

$$\dot{x}(t) = 2\tilde{C}_2 \cos(2t)$$

$$\dot{x}(0) = 1 = 2\tilde{C}_2 \cos(2(0)) = 2\tilde{C}_2 \Rightarrow \tilde{C}_2 = \frac{1}{2}$$

$$\text{Thus, } x(t) = \frac{1}{2} \sin(2t) \quad \checkmark$$

Alternatively: $x(t) = A \cos(2t - \delta)$

$$x(0) = 0 = A \cos(2(0) - \delta) \quad \text{so} \quad \underline{A \cos(\delta) = 0} \quad \begin{array}{l} \swarrow \text{not a typo!} \\ \text{(why not?)} \end{array}$$

$$\dot{x}(t) = -2A \sin(2t - \delta)$$

$$\dot{x}(0) = 1 = -2A \sin(2(0) - \delta) \Rightarrow \underline{2A \sin(\delta) = 1}$$

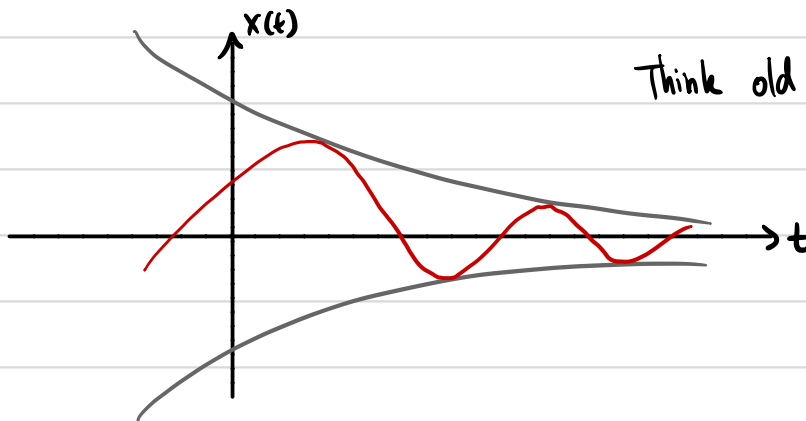
$$A \cos(\delta) = 0 \Rightarrow \delta = \frac{\pi}{2} \quad \text{so, } x(t) = \frac{1}{2} \cos\left(2t - \frac{\pi}{2}\right) = \frac{1}{2} \cos\left(2\left(t - \frac{\pi}{4}\right)\right)$$

$$2A \sin(\delta) = 1 \Rightarrow A = \frac{1}{2}$$

$$\text{or} \quad x(t) = \frac{1}{2} \sin(2t) \quad \text{again!}$$

Sub case (ii): $b \neq 0$. (Underdamped Motion)

$$x(t) = e^{-\frac{b}{2m}t} \left(c_1 e^{\frac{\sqrt{\Delta}}{2m}t} + c_2 e^{-\frac{\sqrt{\Delta}}{2m}t} \right)$$



Think old Caddy suspension.

Since $\Delta < 0$, the terms in the parentheses can be rewritten as

$$x(t) = e^{\alpha t} \left(\tilde{c}_1 \cos(\beta t) + \tilde{c}_2 \sin(\beta t) \right)$$

$$w/ \alpha = -\frac{b}{2m} \text{ and } \beta = \frac{\sqrt{-\Delta}}{2m} = \omega_d = \frac{\sqrt{4mk - b^2}}{2m}$$

Alternatively:

$$x(t) = A(t) \cos(\omega_d t - \delta)$$

$A(t)$ is the time-dependent amplitude $A(t) = A e^{-\frac{b}{2m}t}$

δ phase angle (rad), $\varphi = \delta / \omega_d$

Not entirely periodic:

$\omega_d = \frac{\sqrt{4mk - b^2}}{2m}$ is the circular quasi-frequency (rad/sec)

$f_d = \omega_d / 2\pi$ is the natural quasi-frequency (Hz)

$T = 1/f_d$ is the quasi-period.

$\tau = \frac{2m}{b}$, the time-constant

describe this time! What does it mean? Why is this important?

[It is a measure of your ability as a scientist/engineer to be able to explain this w/ no help]

Ex// $\ddot{x} + \dot{x} + x = 0$, $x(0) = 0$, $\dot{x}(0) = 1$

Characteristic Eqn.: $\lambda^2 + \lambda + 1 = 0$

or $\lambda = \frac{-1 \pm \sqrt{1 - 4(1)(1)}}{2(1)} = -\frac{1}{2} \pm \frac{\sqrt{3}}{2}i$

Thus, $x(t) = e^{-\frac{1}{2}t} \left[c_1 e^{\frac{\sqrt{3}}{2}it} + c_2 e^{-\frac{\sqrt{3}}{2}it} \right]$

$x(t) = e^{-\frac{1}{2}t} \text{ or } \left[\tilde{c}_1 \cos\left(\frac{\sqrt{3}}{2}t\right) + \tilde{c}_2 \sin\left(\frac{\sqrt{3}}{2}t\right) \right]$

$x(0) = 0 \Rightarrow x(0) = 0 = e^{-\frac{1}{2}(0)} \left[\tilde{c}_1 \cos\left(\frac{\sqrt{3}}{2}(0)\right) + \tilde{c}_2 \sin\left(\frac{\sqrt{3}}{2}(0)\right) \right]$
or $\tilde{c}_1 = 0$

$\dot{x}(t) = \tilde{c}_2 \left[\frac{\sqrt{3}}{2} e^{-\frac{1}{2}t} \cos\left(\frac{\sqrt{3}}{2}t\right) - \frac{1}{2} e^{-\frac{1}{2}t} \sin\left(\frac{\sqrt{3}}{2}t\right) \right]$

$\dot{x}(0) = 1 = \tilde{c}_2 \frac{\sqrt{3}}{2} e^0 \cos(0) - \frac{1}{2} e^0 \sin(0) \Rightarrow \tilde{c}_2 = \frac{2}{\sqrt{3}}$

$\therefore x(t) = \frac{2}{\sqrt{3}} e^{-\frac{t}{2}} \sin\left(\frac{\sqrt{3}}{2}t\right) \quad \checkmark$

Case 2, $\Delta = 0$ (critically Damped Motion)

This is a problem! The only solution seems to be $x(t) = C e^{-\frac{b}{2m}t}$!
 $\lambda = \frac{-b \pm \sqrt{0}}{2m} = -\frac{b}{2m}$!

The solution space is 2D, so we need two basis vectors, this yields only one!

Note $\Delta = 0 = b^2 - 4mk$ or $b = 2\sqrt{mk}$

Original ODE: $m\ddot{x} + b\dot{x} + kx = m\ddot{x} + 2\sqrt{mk}\dot{x} + kx \Rightarrow$

$$\left(\sqrt{m} \frac{d}{dt} + \sqrt{k}\right)^2 x = 0$$

$$\left(\sqrt{m} \frac{d}{dt} + \sqrt{k}\right) \underbrace{\left(\sqrt{m} \frac{d}{dt} + \sqrt{k}\right) x}_{\text{call this } y} = 0$$

$$\sqrt{m} \frac{dy}{dt} + \sqrt{k}y = 0 \Rightarrow y = C e^{-\frac{\sqrt{k}}{\sqrt{m}}t}$$

Now $\sqrt{m}\dot{x} + \sqrt{k}x = C e^{-\frac{\sqrt{k}}{\sqrt{m}}t}$

$$\dot{x} + \sqrt{\frac{k}{m}}x = C e^{-\frac{\sqrt{k}}{\sqrt{m}}t}$$

Use Integrating Factor!
 $\mu = e^{\int \frac{\sqrt{k}}{\sqrt{m}} dt} = e^{\frac{\sqrt{k}}{\sqrt{m}}t}$

$$\frac{d}{dt} \left(x e^{\frac{\sqrt{k}}{\sqrt{m}}t} \right) = C$$

$$x e^{\frac{\sqrt{k}}{\sqrt{m}}t} = Ct + D$$

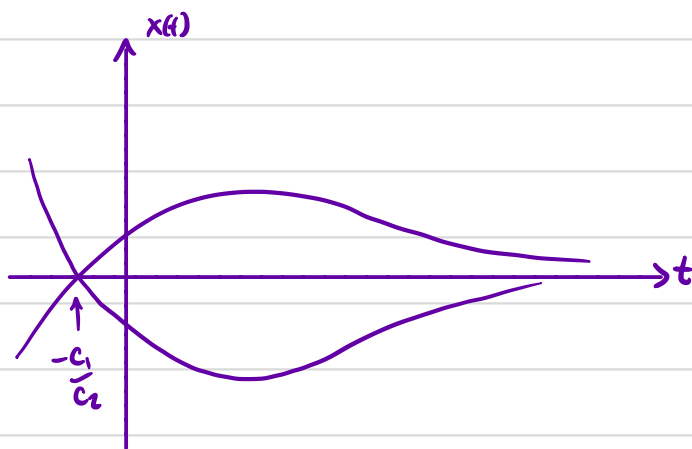
$$x = (Ct + D) e^{-\frac{\sqrt{k}}{\sqrt{m}}t} !$$

Thus, we have two basis vectors: $\{e^{-\sqrt{\frac{k}{m}}t}, te^{-\sqrt{\frac{k}{m}}t}\}$

and $x(t) = \underbrace{c_1 e^{-\frac{b}{2m}t} + c_2 t e^{-\frac{b}{2m}t}}_{\text{Remember } b = 2\sqrt{mk}}$ ✓

Note: For Critically damped motion, the solution has at most one zero!

$$x(t) = 0 = (c_1 + c_2 t)e^{\lambda t} \Rightarrow t = -\frac{c_1}{c_2} \quad \leftarrow \text{root!}$$



Ex/ $\ddot{x} - 4\dot{x} + 4x = 0$, $x(0) = 1$, $\dot{x}(0) = 0$

Characteristic Eqn looks like: $\lambda^2 - 4\lambda + 4 = 0$

$$(\lambda - 2)^2 = 0$$

or $\lambda = 2$ repeated!

$$\therefore x(t) = c_1 e^{2t} + c_2 t e^{2t}$$

$$x(0) = 1 = c_1 e^0 + c_2(0)e^0 \Rightarrow c_1 = 1$$

$$\dot{x}(t) = 2e^{2t} + c_2 [2te^{2t} + e^{2t}]$$

$$\dot{x}(0) = 0 = 2e^0 + c_2 [2(0)e^0 + e^0] = 2 + c_2$$

$$\text{so } c_2 = -2$$

$$x(t) = e^{2t} - 2te^{2t} \quad \checkmark \checkmark$$

Finally, **Case 1: $\Delta > 0$** (overdamped motion)

$$\lambda = \frac{-b \pm \sqrt{b^2 - 4mk}}{2m} \Rightarrow \lambda_1 = \frac{-b + \sqrt{\Delta}}{2m} \quad \lambda_2 = \frac{-b - \sqrt{\Delta}}{2m}$$

Both roots are negative.
(why?)

Complete solution looks like $x(t) = c_1 e^{\lambda_1 t} + c_2 e^{\lambda_2 t}$

Note: Depending on constants (from I.C.'s) the function crosses the t -axis at most once.

Ex// $\ddot{x} + 3\dot{x} + 2x = 0$

The characteristic eqn looks like $\lambda^2 + 3\lambda + 2 = 0$
 $(\lambda + 2)(\lambda + 1) = 0$

$$\lambda = \{-1, -2\}$$

So the solution looks like $x(t) = c_1 e^{-t} + c_2 e^{-2t}$

Suppose $x(0) = 0, \dot{x}(0) = 1,$

$$\begin{aligned} x(0) = 0 &= c_1 + c_2 \\ \dot{x}(0) = 1 &= -c_1 - 2c_2 \end{aligned} \Rightarrow \begin{pmatrix} 1 & 1 \\ -1 & -2 \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$\begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = \frac{1}{(-1)} \begin{pmatrix} -2 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$x(t) = e^{-t} - e^{-2t} \quad \checkmark$$

Summary

$$m\ddot{x} + b\dot{x} + kx = 0 \text{ gives characteristic eqn } m\lambda^2 + b\lambda + k = 0$$

$$\Delta = b^2 - 4mk$$

Case 1 $\Delta > 0$

Two distinct real roots

$$\lambda_1, \lambda_2 = \frac{-b \pm \sqrt{\Delta}}{2m}$$

Overdamped Motion

$$x(t) = c_1 e^{\lambda_1 t} + c_2 e^{\lambda_2 t}$$

Case 2 $\Delta = 0$

One repeated real root

$$\lambda_1 = \lambda_2 = \frac{-b}{2m}$$

Critically Damped Motion

$$x(t) = c_1 e^{\lambda_1 t} + c_2 t e^{\lambda_1 t}$$

Case 3 $\Delta < 0$

Two distinct complex roots

$$\lambda_1, \lambda_2 = \alpha \pm i\beta$$

Underdamped Motion

$$x(t) = e^{\alpha t} (c_1 \cos(\beta t) + c_2 \sin(\beta t))$$

$$\alpha = \frac{-b}{2m} \quad \beta = \frac{\sqrt{4mk - b^2}}{2m}$$